Hammer Drill



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Hammer/drills

What it is used for

A hammer drill rotates a drill bit and provides a hammering action for drilling suitable masonry and other approved materials. Correct bit selection, secure workholding, and controlled operation help produce accurate holes while reducing injury and tool damage.

Main parts

- Motor drives the rotating drill mechanism.
- Chuck holds the drill bit or approved accessory.
- Hammer mechanism produces repeated impact for suitable materials.
- Speed trigger controls the rotation speed.
- Forward and reverse selector changes rotation direction on equipped models.
- Mode selector chooses drilling, hammer drilling, or other supported modes.
- Side handle provides additional grip and control.
- Depth stop limits drilling depth on equipped models.

Before you start

1. Read the manual and confirm the drill is suitable for the material and task.
2. Inspect the body, chuck, side handle, cable or battery, and switches for damage.
3. Select a compatible, undamaged bit of the correct type and size.
4. Check for hidden electrical wires, pipes, and other services before drilling.
5. Secure the workpiece and clear people from the drilling area.
6. Wear safety goggles, hearing protection, and suitable dust protection where required.

How to use

1. Disconnect the power or remove the battery before fitting or changing the bit.
2. Insert the bit fully into the chuck and tighten it according to the manual.
3. Fit the side handle and set the depth stop if required.
4. Select the correct drilling mode and speed for the material.
5. Position the bit at the marked point and hold the drill firmly with both hands.
6. Start at a controlled speed and keep the bit aligned with the intended hole.
7. Apply steady, appropriate pressure without forcing the motor or bit.
8. Withdraw the bit periodically when needed to clear dust and reduce overheating.
9. Release the trigger and wait until the bit stops completely before setting the tool down.
10. Disconnect power or remove the battery before cleaning and inspecting the drill.

⚠ Safety

- Wear eye protection and keep hands away from the rotating bit.
- Never change bits or make adjustments while the drill is powered.
- Do not use damaged bits, loose accessories, or a cracked tool body.
- Keep the side handle fitted where provided and maintain a stable stance.
- Stop immediately if the bit jams, the tool overheats, or unusual vibration occurs.
- Use suitable dust extraction or respiratory protection when drilling materials that generate harmful dust.

Care & storage

- Clean dust from the chuck, vents, and exterior after use.
- Inspect bits for wear, cracks, or deformation and replace damaged bits.
- Check the cord, battery, chuck, side handle, and switches regularly.
- Store the drill and accessories in a dry case according to the manual.

Common mistakes to avoid

- Using a bit unsuitable for the material or drill mode.
- Holding the drill with one hand when firm two-handed control is needed.
- Forcing the drill when the bit binds or progress slows.
- Changing bits while the tool remains connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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